AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 2 · STUDENT LESSON

A Distribution of Possibilities

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 12: probabilities expressed as fractions or decimals sum over outcomes.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Check whether a next-token table is a valid probability distribution. Try the recall before opening notes.

Your learning target

I can check whether a next-token table is a valid **probability distribution**.

I can calculate a two-step **sequence probability** using conditional probabilities.

I can explain why token probability does not measure factual truth.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 2: A **Distribution** of Possibilities

A generator assigns different **likelihood**s to possible **continuation**s.



CREATOR MISSION

Run a weighted-word simulation and compare repeated **output**s.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why must token counts name the vocabulary and special-token convention?

Readiness for Embeddings and position

Do adjacent vocabulary IDs prove that their tokens have similar meanings?

My readiness explanation

This week’s end goal

By the end, I can multiply conditional probabilities to compare paths instead of only first tokens.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied A Distribution of Possibilities case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

> The learners must use a **distribution** of possibilities to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A constructed model uses prefix “Status:” and vocabulary alternatives ready, busy and unknown. Its next-token table is ready 0.5, busy 0.3, unknown 0.2. All entries are nonnegative and sum to 1.

A probability belongs to a token under this prefix and model, not to that token in every situation. An omitted vocabulary alternative would make the interpretation incomplete unless its probability were explicitly zero.

A table containing 0.6,0.3,0.3 sums to 1.2 and is not already a **probability distribution**. If the values are nonnegative weights and the rule says normalise, divide each by 1.2. Do not silently treat weights as probabilities.

Negative weights cannot be repaired by this simple normalisation rule.

Apply the connected idea: Embeddings and position

Use a two-token vocabulary: ID 1 means red with **embedding** [1,0]; ID 2 means blue with **embedding** [0,1]. This lookup table is constructed, not trained.

The **input** red blue has **embedding** rows [1,0] and [0,1], so its shape is two sequence positions by two components. Reversing the text reverses the rows. ID 2 is not “twice as blue” as ID 1 is red.

A model needs a way to use order. A collection of vectors without any order-sensitive procedure may lose that distinction. We introduce one explicit toy scheme: add **position vector** [0,0] at position 0 and [2,0] at position 1.

These invented values teach arithmetic; they are not a recommended production encoding.

STUDENT LESSON · 3 OF 18

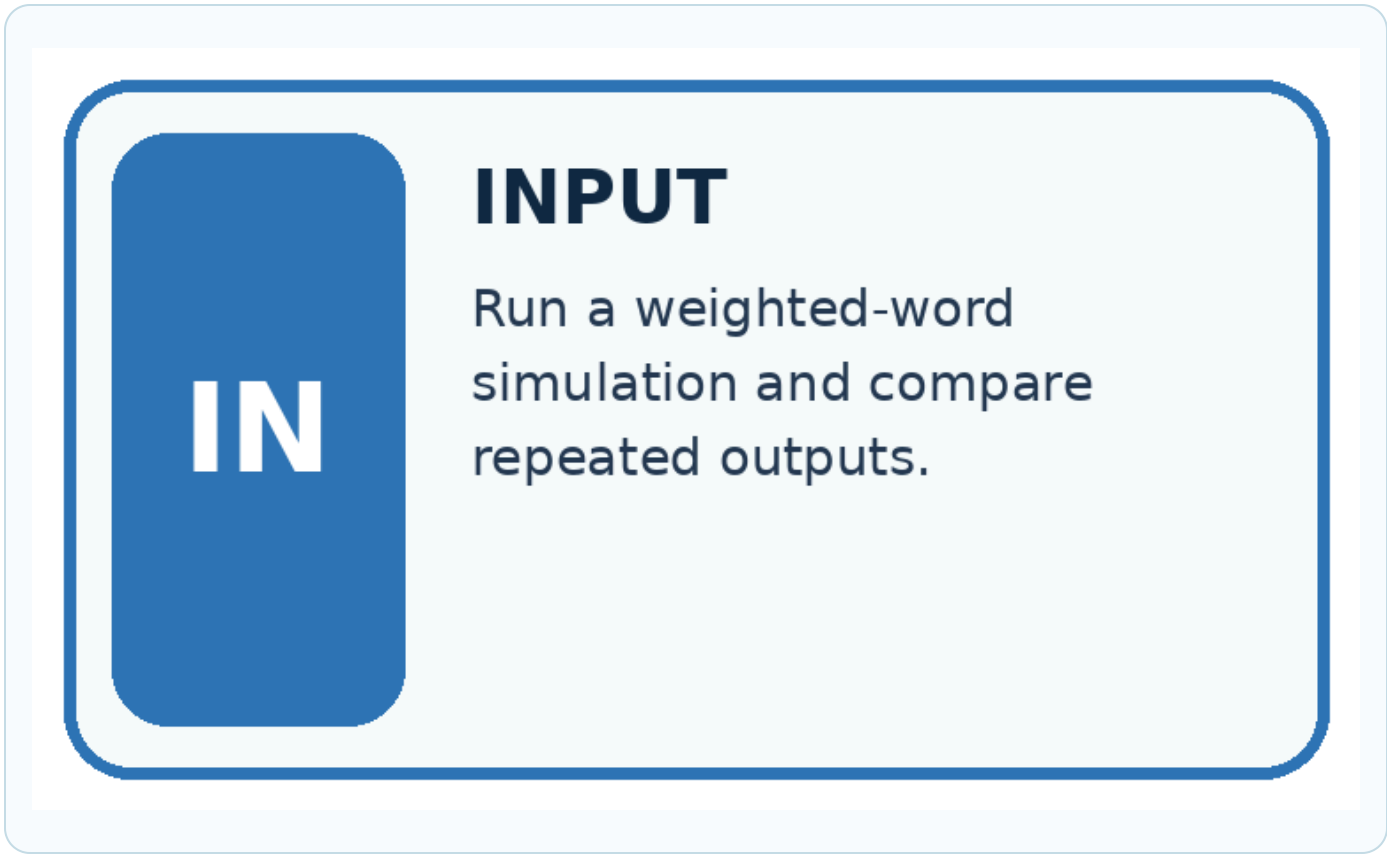
See how it works

> Follow the mechanism for A Distribution of Possibilities. Connect each labelled input, deciding step and result.

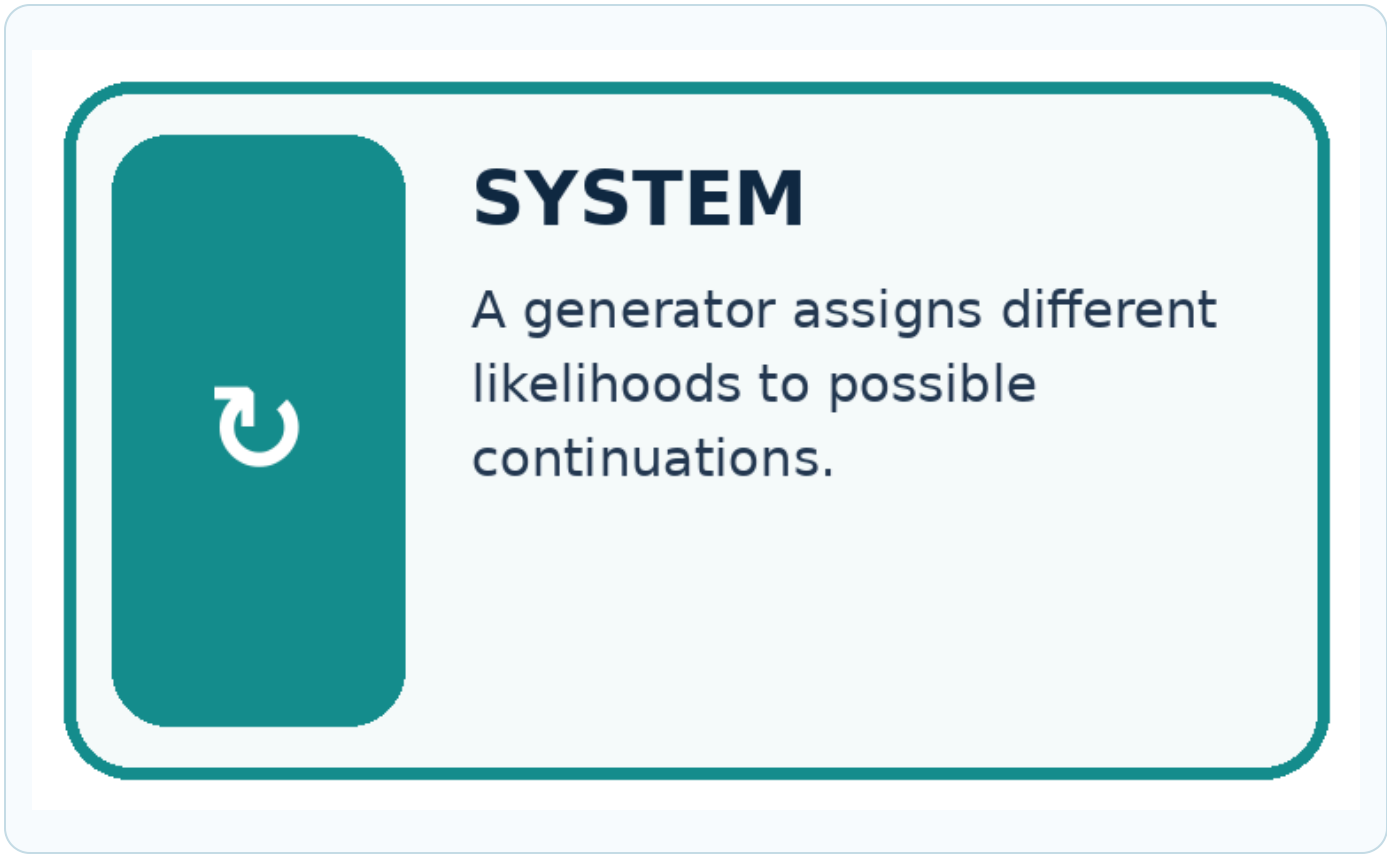
VISUAL MECHANISM

Input → Process → Output

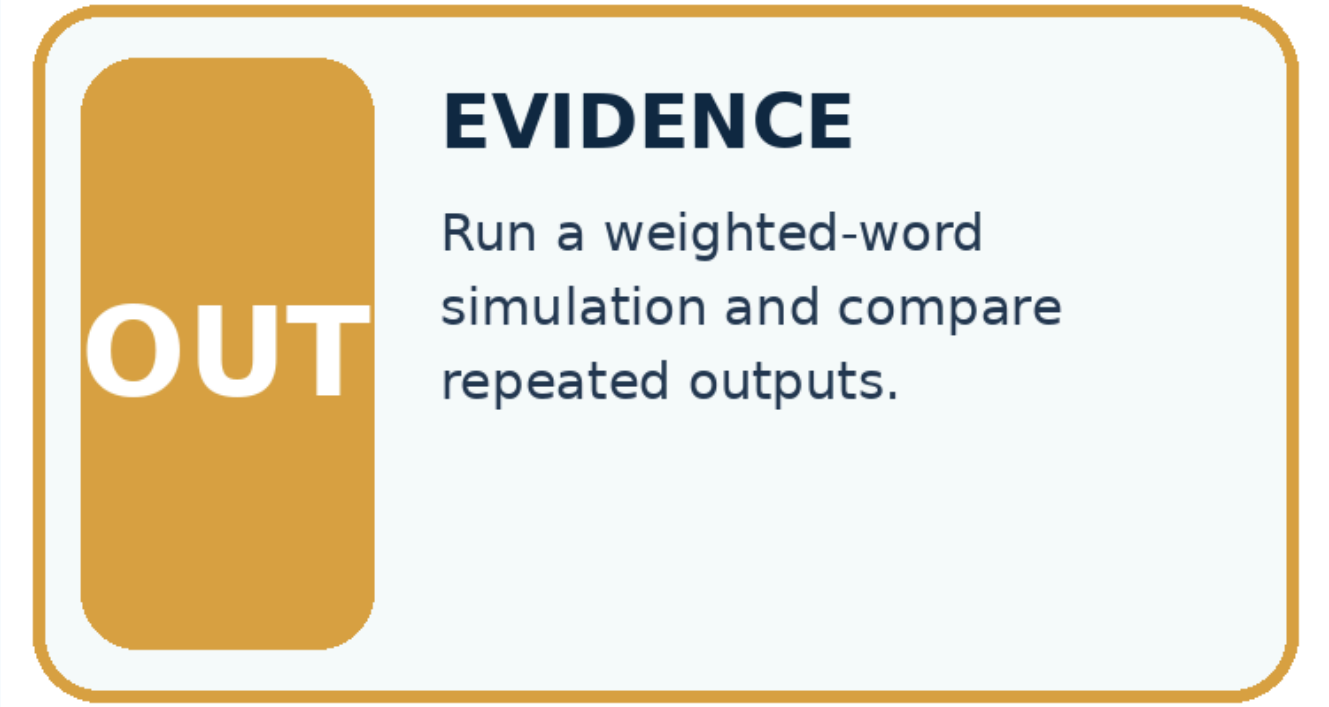
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Run a weighted-word simulation and compare repeated outputs.

↓


SYSTEM
A generator assigns different likelihoods to possible continuations.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Expanded prefix	End probability	Today probability
Status: ready	0.8	0.2
Status: busy	0.5	0.5
Status: unknown	1	0

The probability of ready followed by end is 0.5*0.8 = 0.4. Ready followed by today is 0.5*0.2 = 0.1. Busy followed by end is 0.3*0.5 = 0.15, and busy followed by today is 0.15.

Unknown followed by end is 0.2 and unknown followed by today is 0. Together these six two-step paths sum to 1. These products are chain-rule calculations; they do not assume the second token is independent of the first.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Ready probability 0.5	End given ready 0.8	Joint path probability 0.4
Busy probability 0.3	End given busy 0.5	Joint path probability 0.15
Unknown probability 0.2	End given unknown 1	Joint path probability 0.2

The second probability uses the expanded prefix on that branch.

Apply the connected idea: Embeddings and position

Sequence	Position 0 representation	Position 1 representation
red blue	[1,0] + [0,0] = [1,0]	[0,1] + [2,0] = [2,1]
blue red	[0,1] + [0,0] = [0,1]	[1,0] + [2,0] = [3,0]

The same red token now has [1,0] at position 0 and [3,0] at position 1. Both token identity and position affect the represented row.

Addition requires equal widths; adding a three-component **position vector** to a two-component **embedding** violates the **input** contract. Record the position numbering and **embedding**-table version.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
ID 1 at position 0	[1,0] + [0,0]	Representation [1,0]
ID 1 at position 1	[1,0] + [2,0]	Representation [3,0]
Two different sequences	Inspect ordered rows	Do not infer meaning from ID gaps

The position contribution changes a token representation according to its location.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

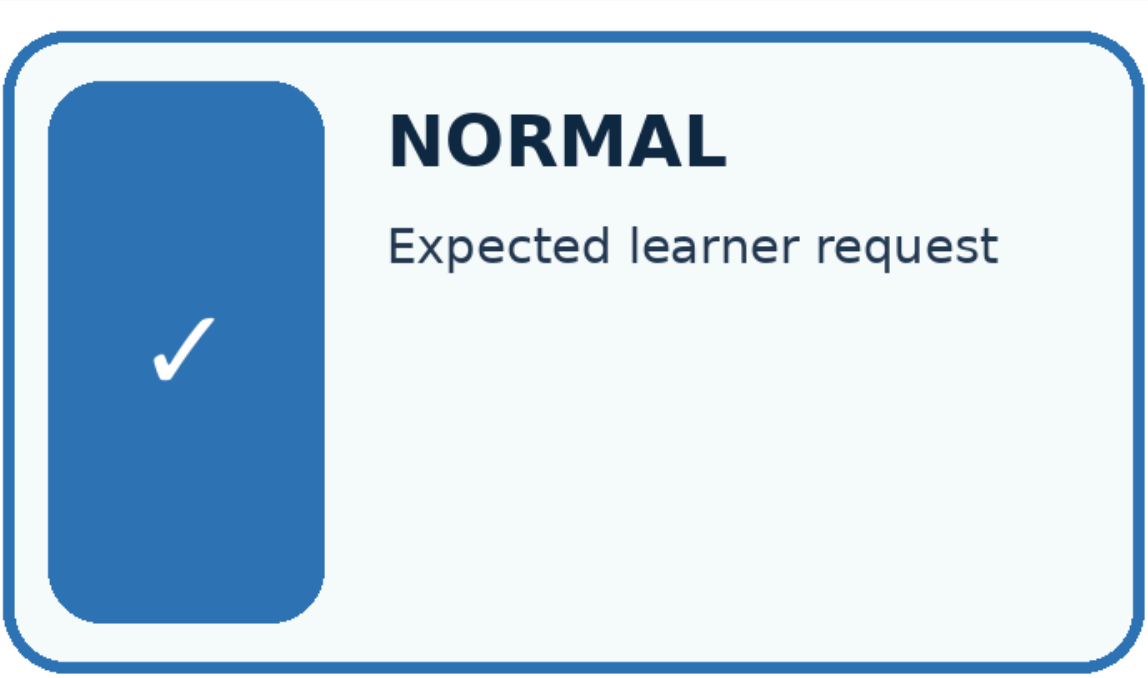
> Compare the supplied cases for A Distribution of Possibilities. Identify the detail that changes the conclusion.

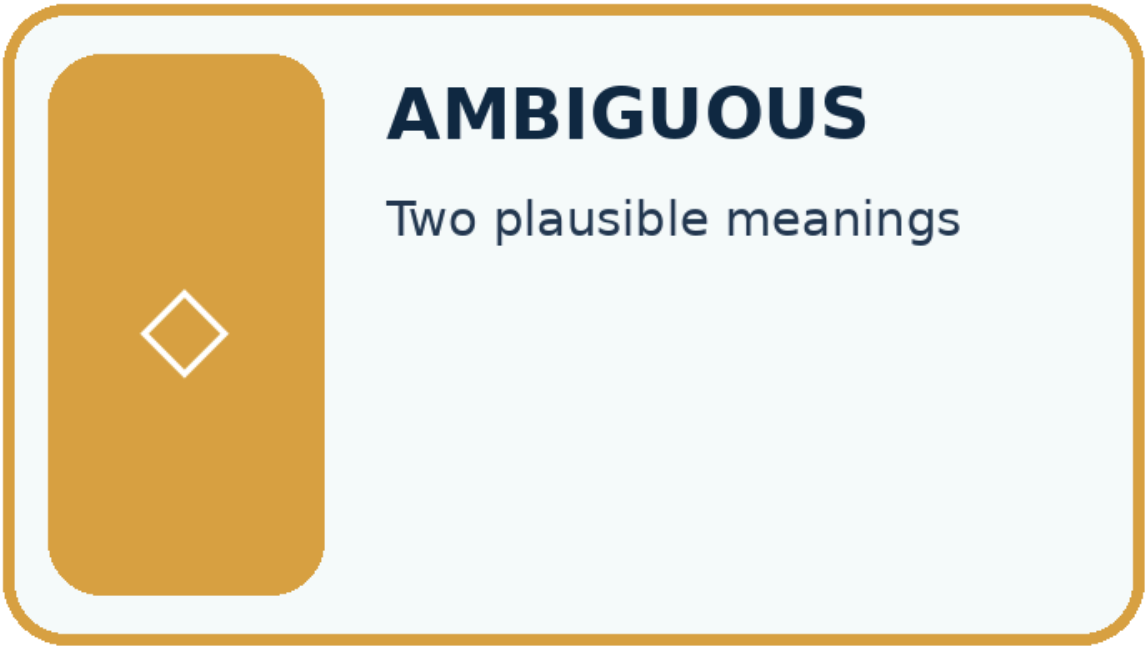
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Predicted result

Prediction: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Why the result follows

Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of A Distribution of Possibilities to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
DISTRIBUTION
possible outcomes with relative likelihoods
2
LIKELIHOOD
relative weight assigned to an outcome





Word	My own example	Not this / boundary
distribution	A description of how values or probabilities are spread across possible outcomes.	
likelihood	In this introductory context, how plausible an outcome is under a model; in statistics, likelihood has a more specific parameter-based meaning.	

Word	My own example	Not this / boundary
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.	
continuation	Text that follows an existing sequence.	

Words used in the closer explanation

Word	Meaning
Probability distribution	Nonnegative probabilities over stated alternatives that sum to one
Conditional probability	A probability evaluated given specified information
Sequence probability	The product of successive conditional probabilities under an autoregressive model
Probability mass	The probability assigned to an alternative or a set of alternatives

Terms for Embeddings and position

Term	Meaning
Embedding	A vector associated with a token under a specified model
Position vector	A vector carrying information about a sequence position
Representation width	The number of components in each vector
Elementwise addition	Adding corresponding vector components

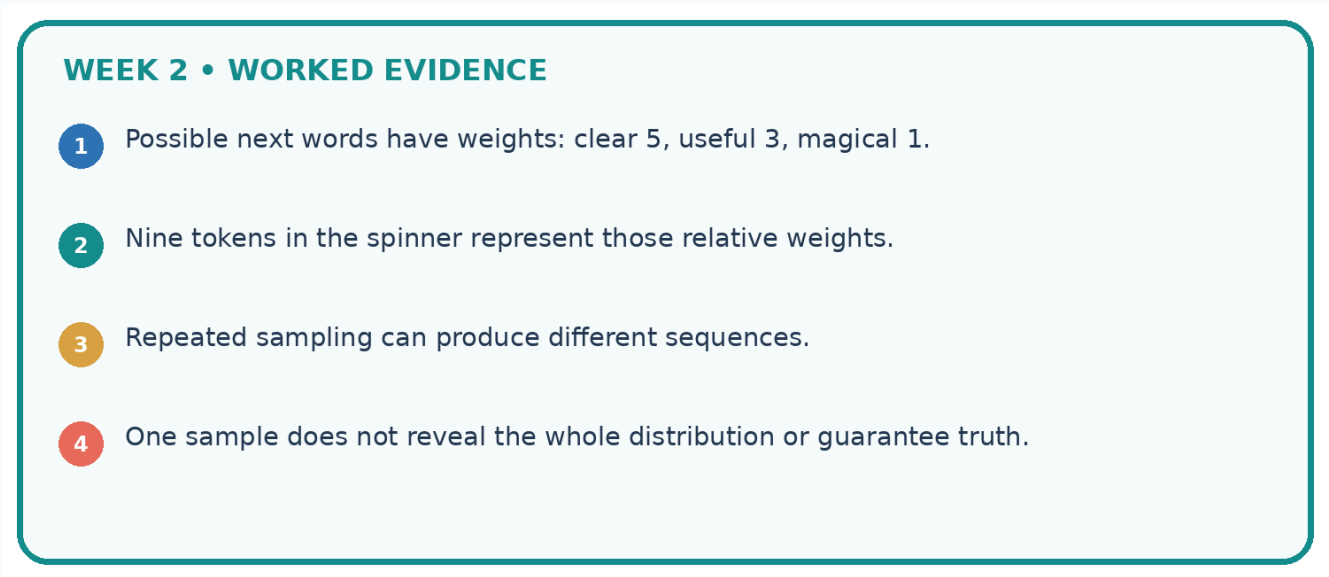
Follow a worked example

> Follow the worked trace for A Distribution of Possibilities, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Possible next words have weights: clear 5, useful 3, magical 1.

2. Nine tokens in the spinner represent those relative weights.

3. Repeated sampling can produce different sequences.

4. One **sample** does not reveal the whole **distribution** or guarantee truth.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · A Distribution of Possibilities

Greedy decoding chooses the largest next-token probability at each step, with an explicit tie rule if needed. It does not always maximise the probability of a complete sequence. Counterexample: first-step A=0.6 and B=0.4.

Given A, end=0.51 and x=0.49; given B, end=1. Greedy produces A,end with probability 0.306. The path B,end has probability 0.4, larger than 0.306.

This comparison concerns exactly two-step paths; longer completion probabilities require the later conditionals too.

A high path probability measures the model's preference over these specified paths. It is not **evidence** that a status claim matches a live **observation**. Format checks and source checks remain separate.

Apply the connected idea: Embeddings and position

Worked explanation · A Distribution of Possibilities

The elementwise sum of the represented rows is [3,1] for both red blue and blue red. This aggregate loses the row-order distinction despite the position additions.

A later computation must actually use the represented rows appropriately; merely adding **position vector**s does not guarantee that every summary preserves order.

Worked explanation · A Distribution of Possibilities

A position scheme may use learned values or a deterministic construction. The original Transformer added positional information to **embedding**s. Modern architectures can use other order-sensitive schemes.

Our two-vector example demonstrates one possible **mechanism** and one information-loss failure, not all architecture choices.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: A Distribution of Possibilities

A fictional two-stage token tree illustrates next-token **prediction**.

The facts we will use

Supplied facts

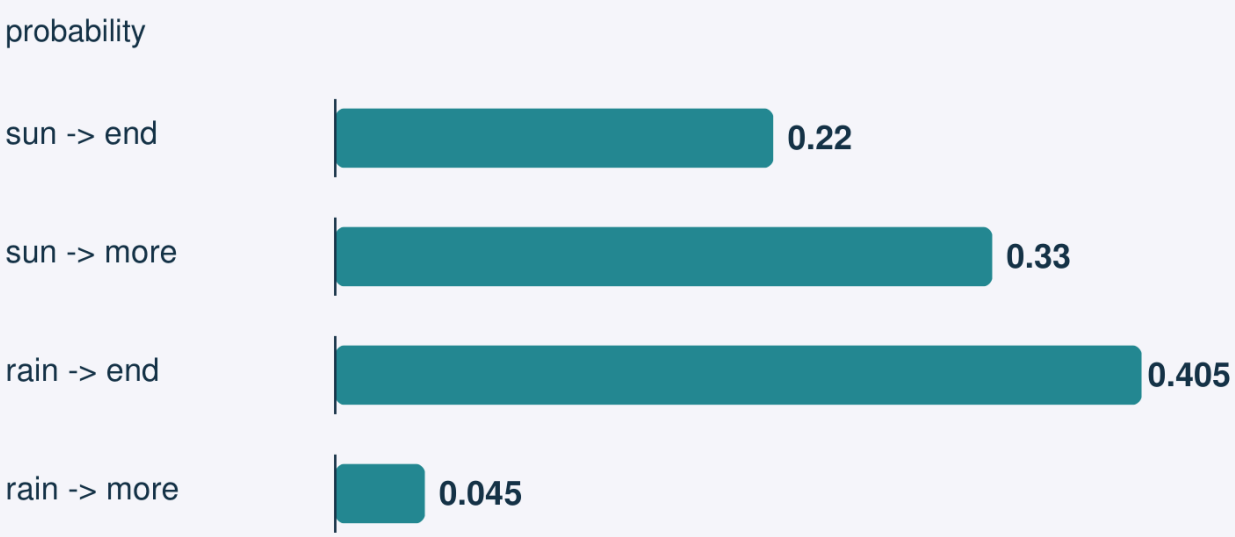
First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

Supplied facts

These are complete toy branches; end paths stop, more paths do not.

AI ACADEMY / AI-3/2

Joint probabilities of the four toy branches



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/2

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end:0.55 x 0.4=0.22; sun -> more:0.55 x 0.6=0.33.



3. Rain branches

rain -> end:0.45 x 0.9=0.405; rain -> more:0.45 x 0.1=0.045.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First distribution	sun 0.55, rain 0.45: greedy first choice is sun.
2. Sun branches	sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33.
3. Rain branches	rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045.
4. Compare end paths	rain→end is the more probable of the two complete end paths.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

What this example does not establish

Keep the stated limit

These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

Keep the stated limit

Read the labelled picture: Why is 0.625 not a broken probability sum?

Keep the stated limit

Change one thing: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the A Distribution of Possibilities claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The model always chooses the single most likely continuation. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Always report the prefix, alternatives and stage. The complement of ready in the first table is busy or unknown, probability 0.5; it is not automatically busy.

Values rounded for display may not sum exactly to 1, so retain full values for calculations and label rounding. A missing or invalid table should lead to an explicit error, not a confident probability claim.

Exit: explain two errors in “ready has probability 0.5, so busy also has probability 0.5 and the room is ready half the time”.

Apply the connected idea: Embeddings and position

Check the claim against the facts

State whether a vector is a token **embedding**, a **position vector** or their combination before calculating distances. Position changes can affect numerical similarity even for the same token.

An unknown token or a sequence beyond the toy position table needs explicit handling. This two-position contract rejects a third position; it does not silently wrap back to position 0.

Exit: a program compares vocabulary IDs instead of vectors and silently repeats its two-position table for a longer sequence. Identify both contract errors.

STUDENT LESSON · 8 OF 18

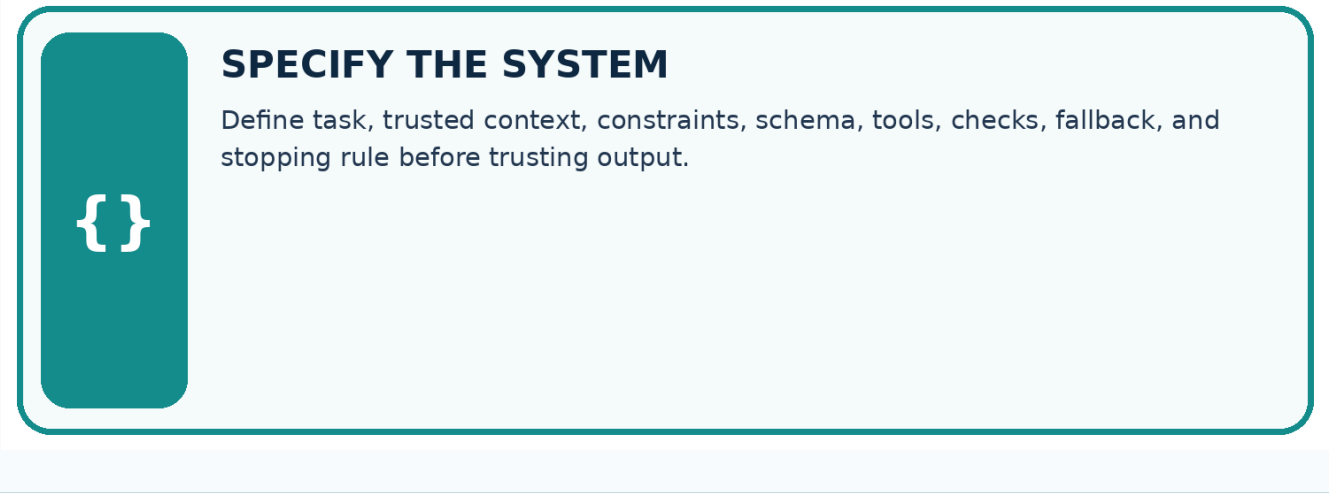
Practise together

Practise A Distribution of Possibilities with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported table: first red=0.4, blue=0.6. End given red=0.9 and end given blue=0.5. The red,end path has probability 0.36. The blue,end path has probability 0.3 despite blue being the larger first-step probability.

Guided: compute both products and distinguish the preferred first token from the more probable specified end path.

Apply the connected idea: Embeddings and position

Supported case: **embedding**s A=[2,1], B=[1,3], positions [0,0] and [1,-1]. Sequence A B becomes [2,1] and [2,2]. Sequence B A becomes [1,3] and [3,0]. Both sums are [4,3].

Guided: compute both ordered representations and compare their sums. Explain which operation discarded the difference.

Practise with less help: Calculate the paths

Use workbook W2 for the connected example “Next token **distribution**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Next token **distribution**s. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Embeddings and position

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A **position vector** guarantees every summary preserves order

Explain the picture

> Why is 0.625 not a broken probability sum?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

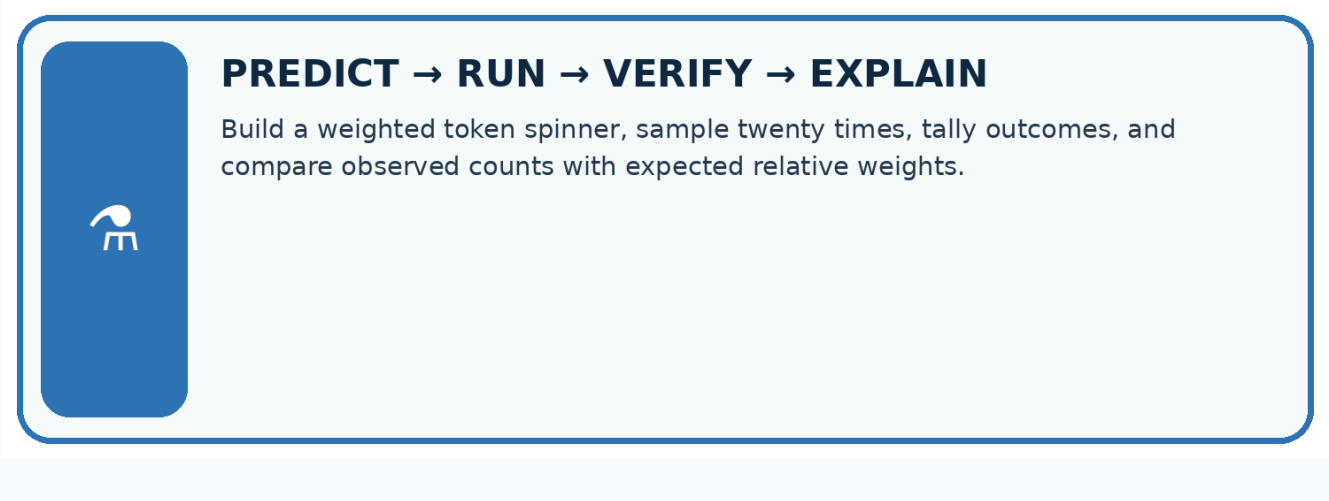
Run the investigation

> Investigate A Distribution of Possibilities. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Build a weighted token spinner, **sample** twenty times, tally outcomes, and compare observed counts with expected relative weights.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

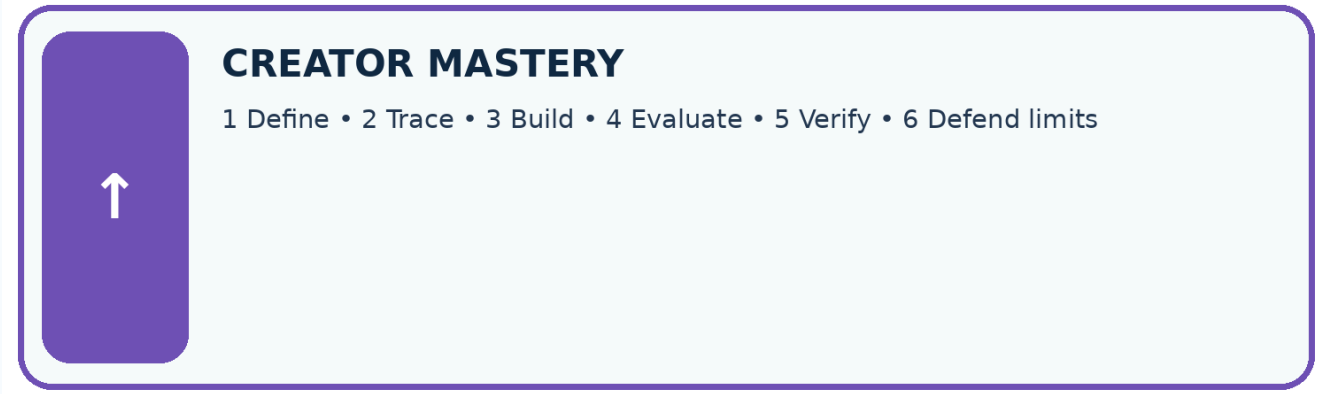
Build your understanding

> Move from recall to tracing, diagnosis and transfer in A Distribution of Possibilities. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

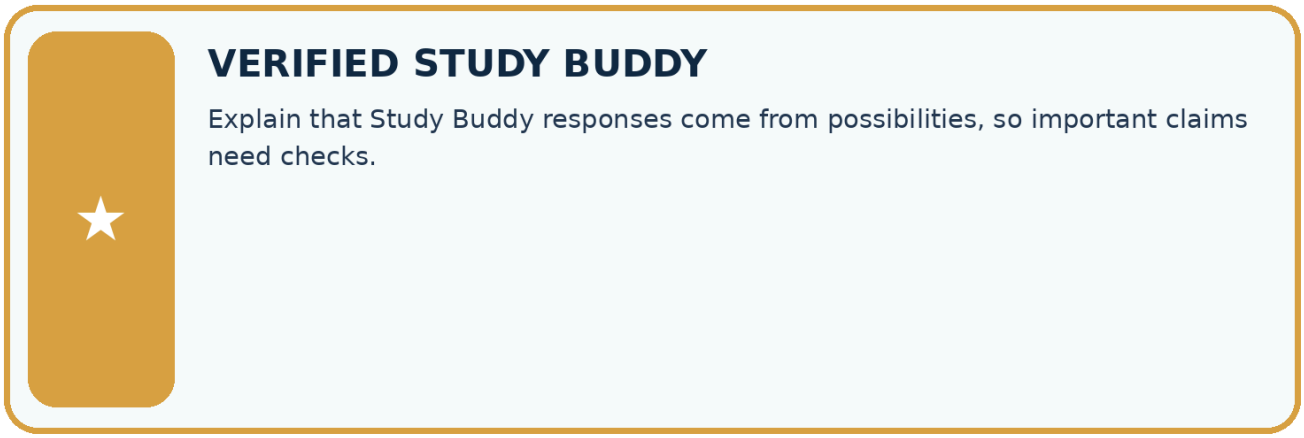
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Explain that Study Buddy responses come from possibilities, so important claims need checks.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

STUDENT LESSON · 12 OF 18

Show what you know

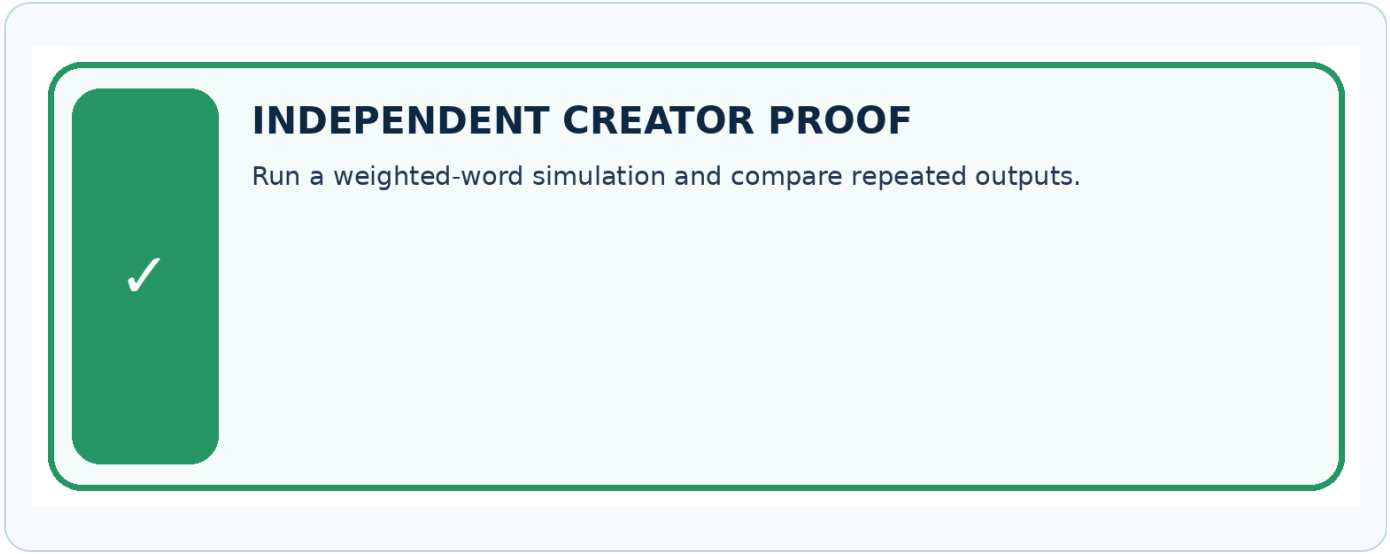
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Run a weighted-word simulation and compare repeated **output**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete A Distribution of Possibilities case for supported practice. Repeated examples do not become fresh assessments.

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A toy next-token **distribution** is rain .5, sun .3, snow .2.

The question

Check total and interpret the largest value.

Worked explanation

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

A toy next-token probability distribution

Evidence or stage	Value or result
rain	50
sun	30
snow	20

A toy next-token **distribution** is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Analyse a new table independently

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A toy next-token **distribution** assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Explain that Study Buddy responses come from possibilities, so important claims need checks.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate A Distribution of Possibilities, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional two-stage token tree illustrates next-token **prediction**.

By the end, I can	How I will show it
Check whether a next-token table is a valid probability distribution	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Calculate a two-step sequence probability using conditional probabilities	Explain the deciding step, show a trace or test, and state one limit.
Explain why token probability does not measure factual truth	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain A Distribution of Possibilities without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Check whether a next-token table is a valid **probability distribution**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional two-stage token tree illustrates next-token **prediction**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? rain→end is the more probable of the two complete end paths.

1. Rain branches

2. Compare end paths

3. Sun branches

4. First **distribution**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare end paths rain→end is the more probable of the two complete end paths. Conditional probabilities describe what follows a particular prefix.

Multiplying along a path gives its joint probability. The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended.

Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. sun→end becomes 0.44, beating unchanged rain→end 0.405.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Repair a missing building block

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

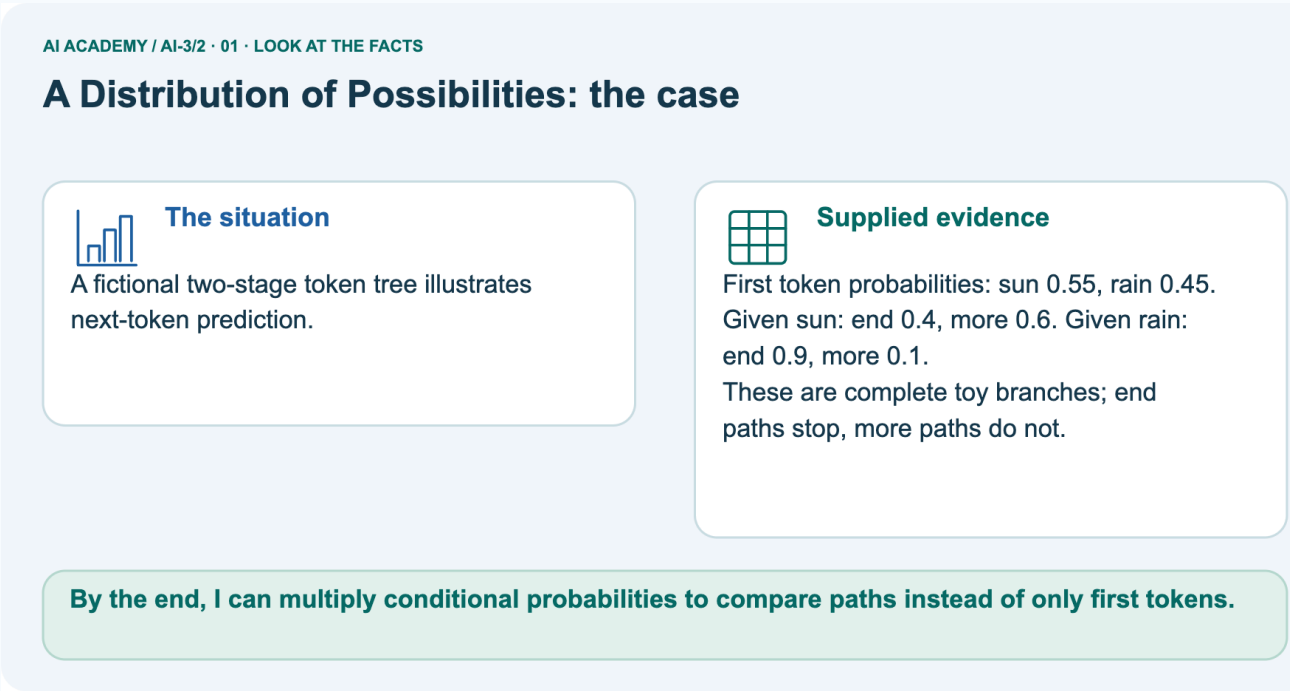
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each A Distribution of Possibilities graphic using its labels, deciding evidence and relevant limitation.

01. A Distribution of Possibilities: the case



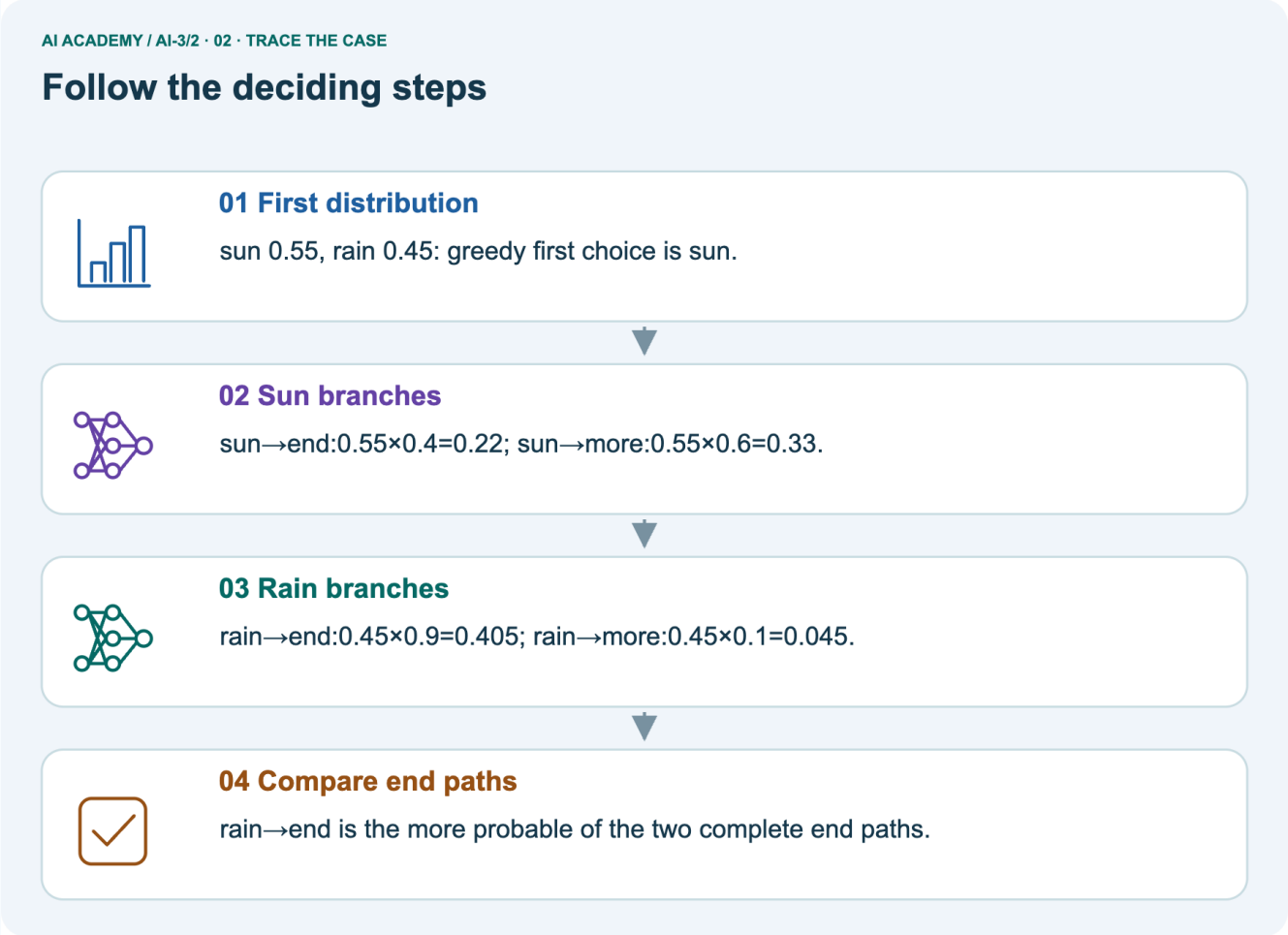
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional two-stage token tree illustrates next-token **prediction**. Supplied facts: First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

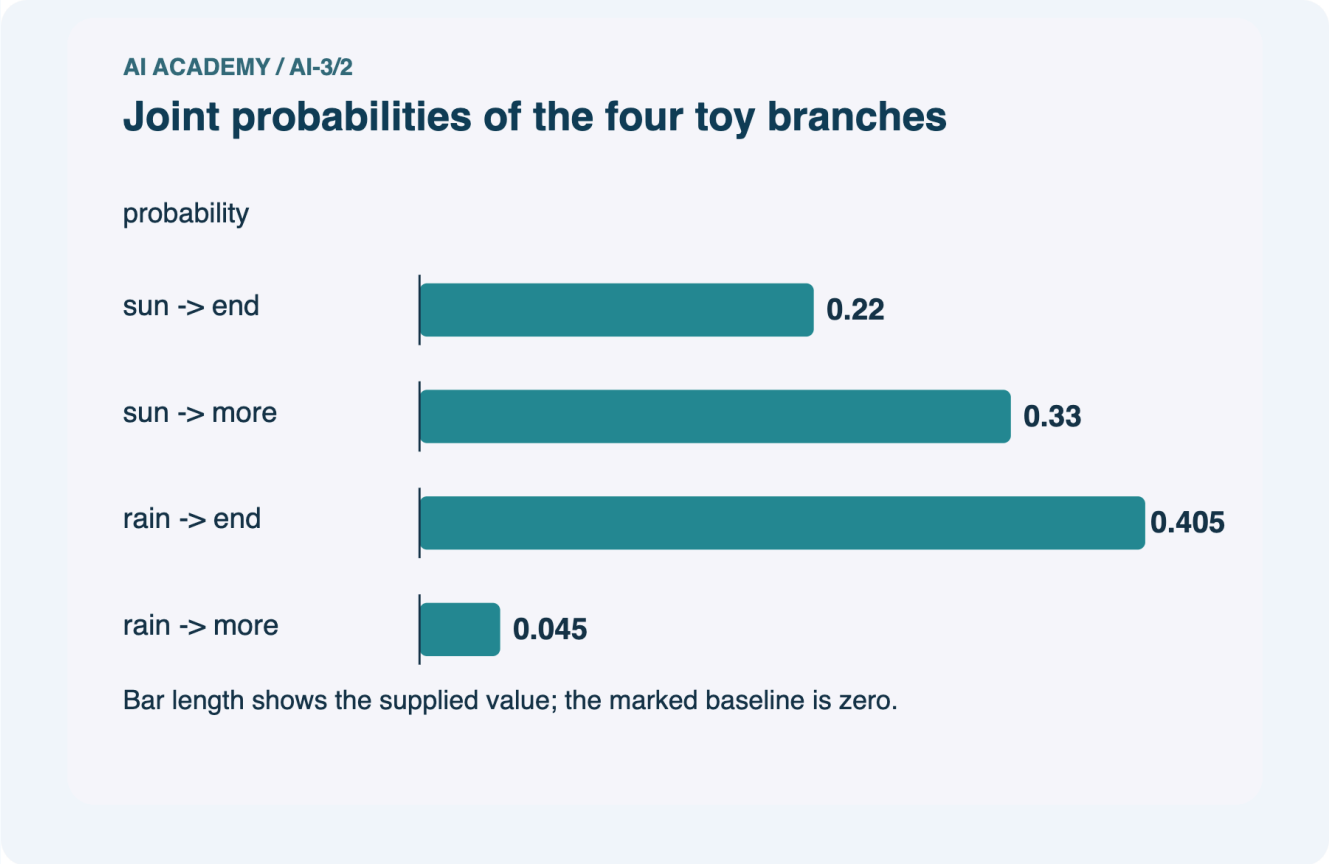


Follow the labelled picture with the matching explanation. Open the vector diagram

First **distribution**: sun 0.55, rain 0.45: greedy first choice is sun. → Sun branches: sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33. → Rain branches: rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045. → Compare end paths: rain→end is the more probable of the two complete end paths.

Point to each numbered stage and explain which recorded **input** it uses.

03. Joint probabilities of the four toy branches

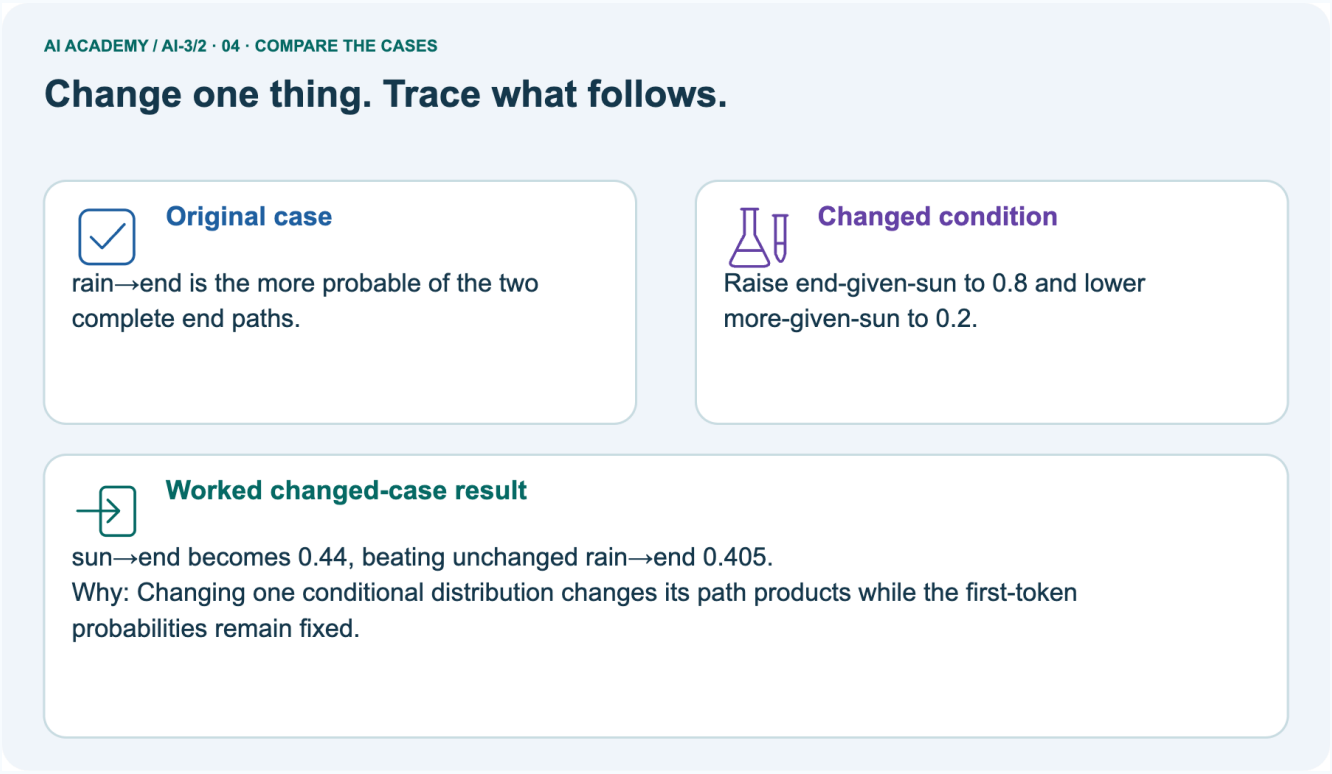


Follow the labelled picture with the matching explanation. Open the vector diagram

Joint probabilities of the four toy branches. Units: probability. sun→end: 0.22. sun→more: 0.33. rain→end: 0.405. rain→more: 0.045.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



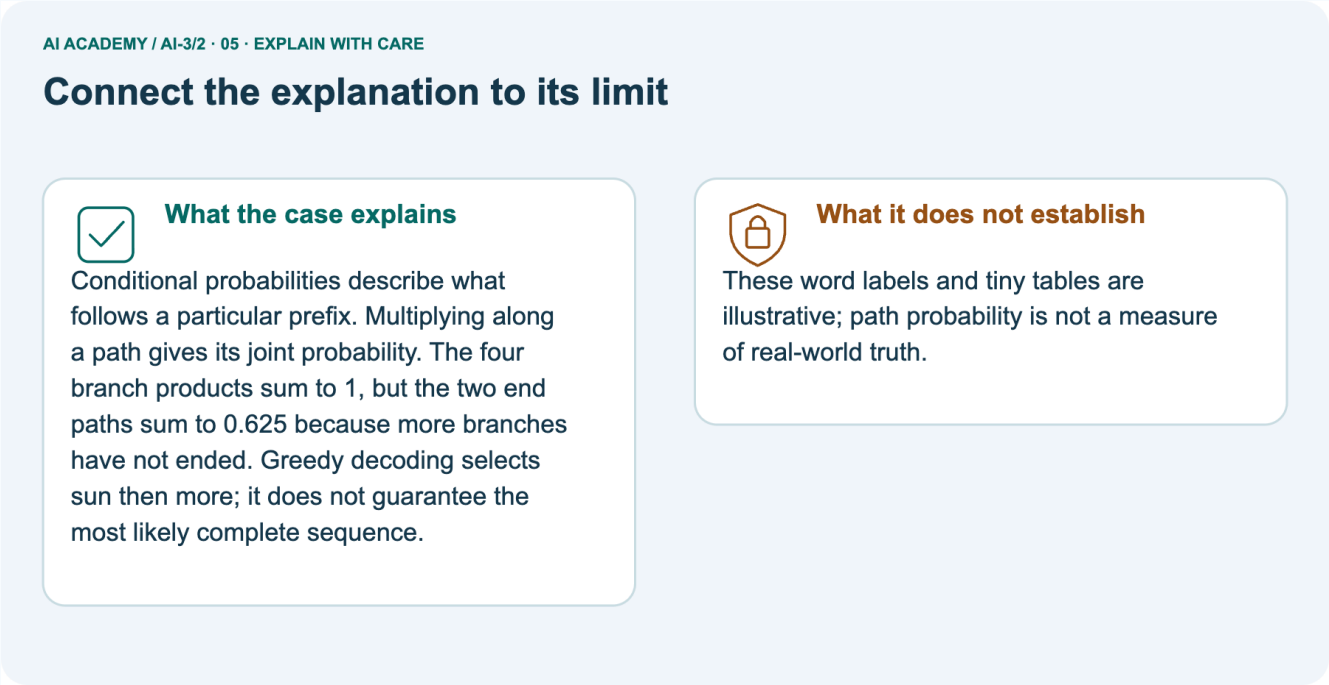
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: rain→end is the more probable of the two complete end paths. Changed condition: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Expected result: sun→end becomes 0.44, beating unchanged rain→end 0.405. Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

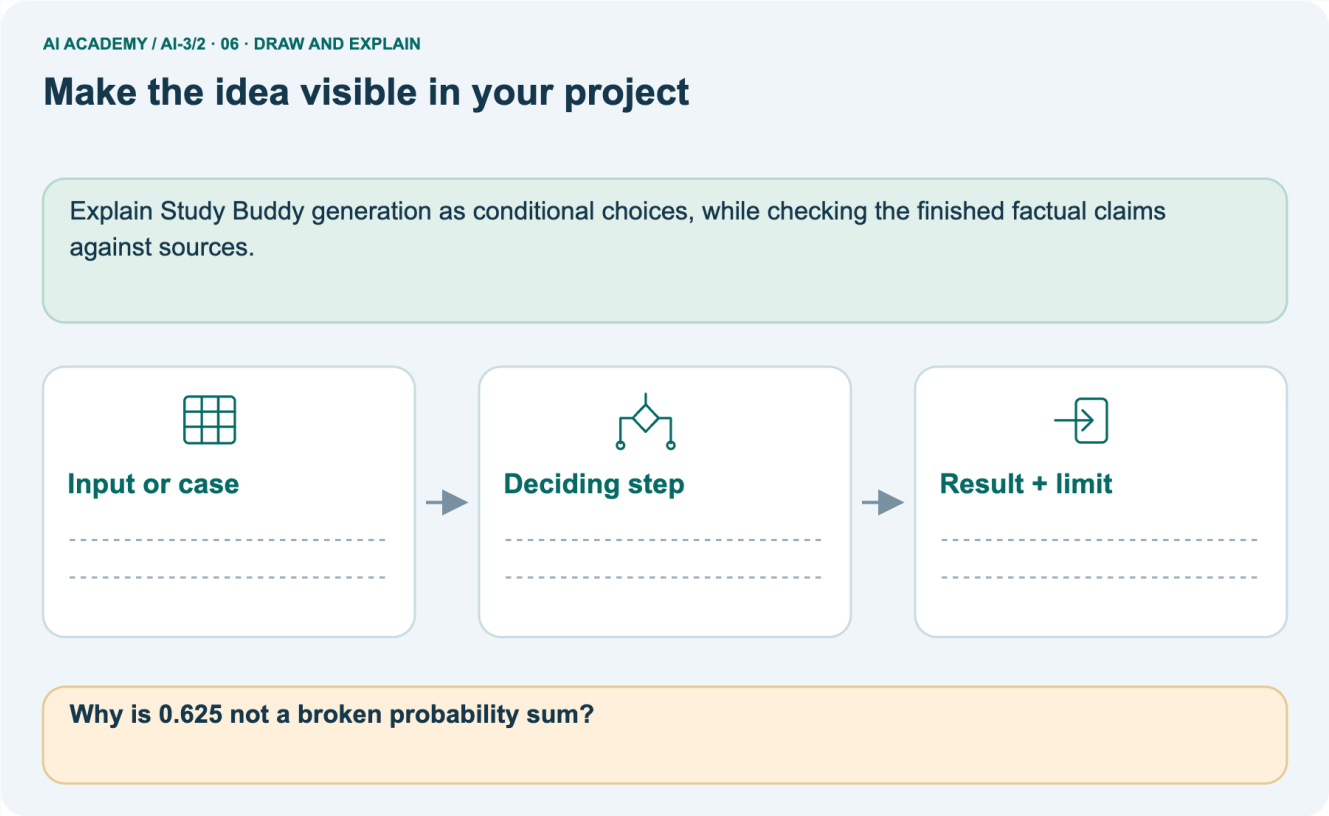
Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Limit: These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources. Draw your **input**, deciding step, result and limit.

Supported question: Why is 0.625 not a broken probability sum?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.